North Carolina EV Charging Infrastructure Analysis

Enhanced Dataset: Connector-Level Detailed Analysis

Dataset: `alt_fuel_stations_ev_charging_units.csv`
Analysis Date: December 29, 2025
Analyst: Advanced Business Analytics Team

Executive Summary

This comprehensive analysis examines North Carolina's electric vehicle charging infrastructure at the **connector-level**, providing unprecedented granularity compared to standard station-level analyses. The dataset contains **1,725 individual charging unit records** from **355 unique stations** across **135 cities** statewide.

🔑 Critical Findings

1. **Tesla NACS Dominance:** 57.5% of all charging units (992 of 1,725) use Tesla's NACS (J3400) standard
2. **High-Power Infrastructure:** 80.5% of units provide ≥150 kW charging capacity
3. **Explosive Recent Growth:** 47.6% of all infrastructure deployed in 2024-2025 alone
4. **Infrastructure Intensity:** Stations average 5.56 connectors, ranging from 1 to 60
5. **Market Concentration:** Top 3 networks (Tesla, ChargePoint, EV Connect) control 59.7% of stations

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Dataset Structure & Methodology

Key Characteristics

Metric	Value
Total Observations	1,725 charging unit records
Unique Stations	355
Geographic Coverage	135 cities across North Carolina
Network Providers	19 distinct charging networks
Temporal Range	2011-2025 (14 years of deployment)

Metric	Value
Data Completeness	>90% for core EV variables

Critical Insight: Row-Level Granularity

This dataset differs fundamentally from typical station-level data:

- Each row represents an **individual charging connector**, not a station
- One station with multiple connector types = multiple rows
- Enables detailed power output and technology adoption analysis
- Average 4.86 connectors per station

Connector Technology Landscape

Connector Type Prevalence

Connector Type	Units with Connector	% of Units	Total Connectors	Avg Power (kW)	Power Range (kW)
NACS (J3400)	992	**57.5%**	1,074	**243.2**	62.0 - 400.0
CCS (J1772COMBO)	638	**37.0%**	638	**215.2**	24.0 - 400.0
CHAdeMO	196	11.4%	196	106.4	24.0 - 360.0
J1772 (Level 2)	65	3.8%	65	15.5	6.7 - 21.6
J3271	0	0.0%	0	—	—

Technology Insights

1. **NACS Ascendancy**
    - 57.5% market penetration demonstrates Tesla's infrastructure leadership
    - 243 kW average power output positions NACS as premium fast-charging standard
    - 717 units at exactly 250 kW (standardized Tesla Supercharger v3)

2. **CCS Maturity**
    - 37% penetration across multiple networks
    - 215 kW average power competitive with NACS
    - Industry standard outside Tesla ecosystem
3. **CHAdeMO Decline**
    - Only 11.4% adoption indicates obsolescence
    - Primarily legacy infrastructure from 2011-2015 Nissan dealerships
    - 106 kW average power lags modern standards
4. **Level 2 Niche**
    - 3.8% of units reflects dataset's DC fast-charging focus
    - 15.5 kW average suitable for workplace/destination charging only

Connector Combinations

Combination	Units	% of Total	Use Case
TESLA	972	56.3%	Tesla Supercharger network
J1772COMBO	322	18.7%	Modern multi-brand DC fast charging
CHADEMO J1772COMBO	252	14.6%	Legacy multi-standard stations
J1772 J1772COMBO	70	4.1%	Mixed Level 2 + DC fast
J1772COMBO TESLA	50	2.9%	Universal compatibility sites

Strategic Implication: Pure Tesla infrastructure (56.3%) vs. multi-standard (43.7%) reflects bifurcated market.

Power Output Analysis

High-Power Infrastructure (≥150 kW)

- **Total High-Power Units:** 1,388 (80.5% of all units)
- **Market Leaders:**
    - Tesla: 940 units (67.7% of high-power infrastructure)

- Electrify America: 110 units
- EV Connect: 77 units
- CIRCLE_K: 49 units

Power Distribution by Connector Type

NACS (J3400):

- Mean: 243.2 kW | Median: 250.0 kW
- Most common: 250 kW (717 units) — Tesla Supercharger v3 standard
- 2nd most common: 150 kW (146 units) — Tesla urban Superchargers
- Range: 62-400 kW

CCS (J1772COMBO):

- Mean: 215.2 kW | Median: 180.0 kW
- Most common: 350 kW (144 units) — Electrify America ultra-fast
- 2nd most common: 180 kW (72 units)
- Wide variability (StdDev: 120.6 kW) reflects diverse deployment strategies

CHAdeMO:

- Mean: 106.4 kW | Median: 100.0 kW
- Most common: 100 kW (53 units), 62.5 kW (46 units)
- Lower average reflects aging infrastructure

J1772 (Level 2):

- Mean: 15.5 kW | Median: 19.2 kW
- Most common: 19.2 kW (18 units) — standard Level 2
- Minimal representation in DC-focused dataset

Business Interpretation

80.5% high-power penetration indicates North Carolina is prioritizing **fast charging** infrastructure over slower Level 2, aligning with interstate corridor and long-distance travel needs.

Network Provider Analysis

Market Share (by Station Count)

Rank	Network	Stations	Market Share	Cumulative %
1	**Tesla**	88	24.79%	24.79%
2	**ChargePoint Network**	77	21.69%	46.48%
3	**EV Connect**	47	13.24%	59.72%
4	eVgo Network	21	5.92%	65.64%
5	Non-Networked	21	5.92%	71.56%
6	Electrify America	20	5.63%	77.19%
7	Blink Network	16	4.51%	81.70%
8	FORD_CHARGE	16	4.51%	86.21%
9	CIRCLE_K	11	3.10%	89.31%
10	IONNA	6	1.69%	91.00%

Total Charging Capacity by Network

Network	Total Connectors	J1772	CCS	CHAdeMO	NACS
Tesla	**1,054**	0	0	0	**1,054**
ChargePoint Network	153	0	138	13	2
EV Connect	153	5	125	23	0
Electrify America	116	1	110	5	0
eVgo Network	110	0	90	20	0

Infrastructure Intensity

Network	Avg Connectors/Station	Total Connectors	Station Count	Strategy
Tesla	**11.98**	1,054	88	**Hub Model**
Electrify America	5.80	116	20	Mid-Capacity Hubs
IONNA	8.00	48	6	Large Hubs
eVgo Network	5.24	110	21	Mid-Capacity
CIRCLE_K	5.27	58	11	Convenience Store Model
EV Connect	3.26	153	47	**Distribution Model**
FORD_CHARGE	3.50	56	16	Moderate Distribution
Blink Network	3.00	48	16	Distribution
Non-Networked	2.81	59	21	Small Dealership Sites
ChargePoint Network	1.99	153	77	**Wide Distribution**

Strategic Business Models

Hub Strategy (High Connectors/Station):

- **Tesla (11.98 avg):** Few locations, high capacity
- **Electrify America (5.80 avg):** Interstate corridor hubs
- **IONNA (8.00 avg):** Automaker-backed premium hubs

Distribution Strategy (Low Connectors/Station):

- **ChargePoint (1.99 avg):** Maximum geographic reach
- **EV Connect (3.26 avg):** Urban/suburban distribution

Geographic Distribution

Top 20 Cities by Station Count

Rank	City	Stations	% of State	Cumulative %
1	**Charlotte**	21	5.92%	5.92%
2	Greensboro	13	3.66%	9.58%
3	Raleigh	12	3.38%	12.96%
4	Asheville	12	3.38%	16.34%
5	Durham	11	3.10%	19.44%
6	Cary	10	2.82%	22.26%
7	Wilmington	10	2.82%	25.08%
8	Fayetteville	7	1.97%	27.05%
9	Hickory	7	1.97%	29.02%
10	New Bern	7	1.97%	30.99%

Geographic Insights

- **Well-Distributed Infrastructure:** Top 5 cities = only 19.4% of stations
- **135 cities served** demonstrates strong rural/small-town coverage
- **Charlotte metro dominance:** 21 stations (5.92%) aligns with population
- **Research Triangle:** Raleigh (12) + Durham (11) + Cary (10) = 33 stations (9.3%)
- **Tourism hubs:** Strong presence in Asheville (12) and Wilmington (10)

Deployment Timeline

Stations Opened by Year

Year	Stations Opened	% of Total	Cumulative Stations	Cumulative %
2011	4	1.1%	4	1.1%
2012	11	3.1%	15	4.2%
2018	7	2.0%	31	8.7%
2019	10	2.8%	41	11.5%
2020	18	5.1%	59	16.6%
2021	38	10.7%	97	27.3%
2022	30	8.5%	127	35.8%
2023	**59**	**16.6%**	**186**	**52.4%**
2024	**81**	**22.8%**	**267**	**75.2%**
2025	**88**	**24.8%**	**355**	**100.0%**

Growth Analysis

- **2024-2025: 169 stations (47.6% of total infrastructure)**
- **2020-2025: 314 stations (88.5% of total infrastructure)**
- **2021 Inflection Point:** Federal EV incentives drove acceleration from 18 (2020) → 38 (2021) stations
- **Sustained Growth:** Every year 2020-2025 exceeded previous totals

Year-over-Year Growth Rates (2020-2025)

Period	Growth Rate
2020→2021	+111.1%
2021→2022	-21.1% (consolidation)
2022→2023	+96.7%

Period	Growth Rate
2023→2024	+37.3%
2024→2025	+8.6% (deceleration)

Interpretation: Growth is decelerating after explosive 2023-2024, suggesting market maturation.

Station Size Distribution

Size Category	Stations	Percentage	Avg Use Case
Micro (1-2 connectors)	142	40.0%	Dealerships, small businesses
Small (3-5 connectors)	80	22.5%	Gas stations, convenience stores
Medium (6-10 connectors)	77	21.7%	Shopping centers, highway rest stops
Large (11-20 connectors)	52	14.6%	Major Tesla Superchargers, travel plazas
Mega (20+ connectors)	4	1.1%	Flagship Tesla Superchargers

Station Size Statistics

- **Mean:** 5.56 connectors per station
- **Median:** 4.0 connectors per station
- **Range:** 1 - 60 connectors
- **Std Dev:** 5.22 (high variability)

Business Insight: 40% of stations are "micro" (1-2 connectors), indicating many small, distributed sites rather than consolidated hubs. This contrasts with Tesla's hub strategy.

Facility Type & Access Analysis

Facility Type Distribution (38.6% data completeness)

Facility Type	Stations	% of Known
PUBLIC	27	19.7%
SHOPPING_CENTER	26	19.0%
CAR_DEALER	19	13.9%
GAS_STATION	17	12.4%
CONVENIENCE_STORE	13	9.5%
RESTAURANT	10	7.3%
GROCERY	8	5.8%
PARKING_LOT	7	5.1%
HOTEL	4	2.9%

Retail Integration Analysis

60% of stations at retail locations (shopping, gas, convenience, grocery):

- Enables revenue from foot traffic during 15-30 min charging sessions
- Co-location with complementary businesses (coffee, dining) drives adoption
- Real estate partnerships crucial to network expansion

Access Type

- **100% Public Access** across all 355 stations
- No workplace-only or private access restrictions
- Demonstrates NC's commitment to universal charging accessibility

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Tesla NACS (J3400) Adoption

NACS Deployment Statistics

- **Total NACS Units:** 992 (57.5% of all charging units)
- **Networks Using NACS:**
  - Tesla: 972 units (97.98% of NACS)
  - IONNA: 18 units (1.81%)
  - ChargePoint Network: 2 units (0.20%)

NACS Power Output

- **Mean:** 243.2 kW
- **Median:** 250.0 kW
- **Range:** 62 - 400 kW
- **Most Common:** 250 kW (717 units) — Tesla Supercharger v3
- **2nd Most Common:** 150 kW (146 units) — Urban Superchargers

Strategic Implications

1. **Market Dominance:** 57.5% NACS penetration indicates Tesla's infrastructure hegemony
2. **Emerging Adoption:** IONNA's 18 NACS units signal automaker coalition embracing Tesla standard
3. **Power Advantage:** 243 kW average positions NACS as premium fast-charging technology
4. **Standardization Trend:** ChargePoint's 2 NACS units suggest tentative multi-network adoption

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Connector Strategy by Network

Network-Specific Technology Choices

Network	J1772 %	CCS %	CHAdeMO %	NACS %	Strategy
Tesla	0%	0%	0%	**100%**	Pure NACS proprietary
ChargePoint Network	0%	100%	89.6%	2.6%	Multi-standard (legacy CHAdeMO)
EV Connect	10.6%	100%	53.2%	0%	Multi-standard focus

Network	J1772 %	CCS %	CHAdeMO %	NACS %	Strategy
eVgo Network	0%	100%	95.2%	0%	CCS + CHAdeMO
Non-Networked	**95.2%**	52.4%	**95.2%**	0%	**Legacy dealerships**
Electrify America	5%	100%	25%	0%	CCS-first, selective CHAdeMO
Blink Network	25%	100%	31.2%	0%	Mixed strategies
FORD_CHARGE	**0%**	**100%**	**0%**	**0%**	**Pure CCS**

Strategic Insights

1. **Tesla Proprietary Lock-in:** 100% NACS deployment maintains ecosystem control
2. **Legacy CHAdeMO Burden:** ChargePoint (89.6%), eVgo (95.2%), and Non-Networked (95.2%) maintain expensive legacy infrastructure
3. **CCS Universal Adoption:** 100% CCS presence among modern networks (ChargePoint, EV Connect, eVgo, Electrify America, Ford)
4. **Automaker Alignment:** Ford's pure CCS strategy reflects industry standard outside Tesla

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Business Insights & Strategic Recommendations

INSIGHT 1: Tesla's Infrastructure Moat

Finding:

- 24.8% of stations, but **60.5% of total connectors** (1,054 of 1,740)
- 11.98 avg connectors/station vs. 1.99 for ChargePoint
- 100% NACS proprietary standard creates switching costs

Implications:

- Tesla owners have **5.9x more charging options** (1,054 NACS vs. 177 multi-brand DC fast)
- Network effects reinforce Tesla vehicle sales
- Competitors must overbuild to match Tesla's effective coverage

Recommendations:

1. **For Non-Tesla Networks:** Accelerate NACS adapter deployment to access Tesla's 1,054 connectors
2. **For Tesla:** Monetize Supercharger network via third-party access (already begun with select automakers)
3. **For Policy Makers:** Require interoperability standards in NEVI funding to prevent proprietary lock-in

INSIGHT 2: High-Power Infrastructure Dominance

Finding:

- 80.5% of units provide ≥150 kW charging
- Tesla NACS: 243 kW avg | CCS: 215 kW avg
- Only 3.8% Level 2 representation

Implications:

- NC prioritizes **interstate corridor** and **long-distance travel** over destination/workplace charging
- Fast charging enables "gas station replacement" model (15-30 min refuel)
- Level 2 infrastructure underrepresented in this dataset (likely exists separately)

Recommendations:

1. **For Businesses:** Co-locate amenities (food, retail) with DC fast charging for 20-30 min dwell time monetization
2. **For Urban Planners:** Balance DC fast corridor infrastructure with Level 2 residential/workplace charging
3. **For Utilities:** Design tariffs that encourage off-peak DC fast charging to manage grid load

INSIGHT 3: Explosive Recent Growth with Deceleration Signals

Finding:

- 169 stations (47.6%) opened in 2024-2025
- Growth rate declining: 37.3% (2023→2024) → 8.6% (2024→2025)
- 2021 inflection point driven by federal EV incentives

Implications:

- Market transitioning from exponential growth to S-curve maturation

- Federal incentives (NEVI, IRA) successfully catalyzed deployment
- Deceleration suggests approaching short-term supply-demand equilibrium

Recommendations:

1. **For Investors:** Focus on **utilization metrics** and **profitability**, not just station counts
2. **For Network Operators:** Optimize existing sites (add chargers) rather than endless expansion
3. **For Policy Makers:** Shift from deployment incentives to **utilization incentives** (e.g., subsidize low-traffic rural sites)

INSIGHT 4: Geographic Distribution vs. Concentration

Finding:

- 135 cities served (excellent rural penetration)
- Top 5 cities = only 19.4% of stations (low concentration)
- Charlotte (21), Greensboro (13), Raleigh (12) lead

Implications:

- NC avoided over-concentration in major metros (unlike CA, NY)
- Strong rural/interstate corridor coverage
- Risk: Low utilization in small towns may threaten economic viability

Recommendations:

1. **For Network Operators:** Monitor utilization in low-traffic rural sites; consider consolidation
2. **For Tourism Boards:** Market EV-friendly corridors (Outer Banks, Blue Ridge Parkway) to attract EV tourism
3. **For State Government:** Protect rural charging access via targeted subsidies or public-private partnerships

INSIGHT 5: Connector Standard Fragmentation

Finding:

- 4 major standards: NACS (57.5%), CCS (37%), CHAdeMO (11.4%), J1772 (3.8%)

- Legacy CHAdeMO drags on network economics (maintenance, low utilization)
- Tesla vs. everyone bifurcation

Implications:

- Consumer confusion: "Which charger works with my car?"
- Stranded asset risk for CHAdeMO infrastructure
- NACS adoption by Ford, GM, Rivian may shift landscape 2025-2027

Recommendations:

1. **For Networks:** Phase out CHAdeMO, focus on CCS + NACS
2. **For Automakers:** Accelerate NACS adoption to simplify consumer experience
3. **For Consumers:** Prioritize vehicles with CCS or NACS (avoid CHAdeMO-only vehicles)

INSIGHT 6: Infrastructure Intensity Business Models

Finding:

- Hub Model (Tesla 11.98 avg, IONNA 8.0 avg): Few locations, high capacity
- Distribution Model (ChargePoint 1.99 avg, EV Connect 3.26 avg): Many locations, low capacity
- Tesla's 1,054 connectors from 88 stations vs. ChargePoint's 153 from 77 stations

Implications:

- Hub Model: Better for long-distance travel, higher utilization per site
- Distribution Model: Better geographic accessibility, lower per-site utilization
- No clear winner; strategies serve different customer segments

Recommendations:

1. **For High-Traffic Corridors (I-85, I-40):** Hub model with 10-15 chargers per site
2. **For Urban/Suburban:** Distribution model with 2-4 chargers per site
3. **For Rural:** Hybrid approach with mid-capacity sites (5-7 chargers) at strategic intervals

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Comparison with Previous Analysis

Dataset Differences

Metric	Previous Dataset	Enhanced Dataset	Difference
Total Rows	355	1,725	+386%
Granularity	Station-level	Connector-level	More detailed
Connector Details	Basic counts	Power outputs, types	Actionable insights
Power Analysis	Limited	Comprehensive	Strategic advantage

Key New Insights from Enhanced Dataset

1. **Power Output Distribution:** Enhanced dataset reveals 243 kW avg for NACS vs. 215 kW for CCS
2. **Tesla's True Dominance:** 60.5% of connectors (not just 24.8% of stations) clarifies market position
3. **High-Power Focus:** 80.5% ≥150 kW quantifies NC's fast-charging priority
4. **CHAdeMO Obsolescence:** 11.4% penetration quantifies legacy standard decline
5. **Station Size Variability:** 1-60 connector range reveals diverse deployment strategies

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Data Quality Assessment

Strengths

- ✅ Perfect completeness for core EV variables (connector types, power outputs)
- ✅ Geographic data (lat/long) 100% complete
- ✅ Temporal data (Open Date) 100% complete
- ✅ Network assignment 100% complete

Limitations

- ⚠️ Facility Type: Only 38.6% complete (137 of 355 stations)
- ⚠️ EV Pricing: Only 11% complete (limits competitive analysis)

- ⚠️ Level 2 data: Severely underreported (likely dataset bias toward DC fast charging)
- ⚠️ 2025 data: May be incomplete (88 stations reported for full year)

Recommendations for Future Data Collection

1. Mandate facility type reporting for all NEVI-funded stations
2. Require standardized pricing transparency
3. Separate Level 2 infrastructure tracking
4. Real-time utilization data integration

Conclusions

Summary of Key Findings

North Carolina's EV charging infrastructure exhibits:

✅ **Explosive Growth:** 47.6% of infrastructure deployed in 2024-2025
✅ **Tesla Dominance:** 60.5% of connectors, 100% NACS proprietary standard
✅ **High-Power Focus:** 80.5% of units ≥150 kW (interstate corridor priority)
✅ **Geographic Distribution:** 135 cities served (strong rural coverage)
✅ **Market Concentration:** Top 3 networks = 59.7% of stations

⚠️ **Challenges:**

- CHAdeMO legacy burden (11.4% of units, declining standard)
- Growth deceleration (37% → 8.6% YoY)
- Connector standard fragmentation (4 major types)
- Facility type data gaps (38.6% completeness)

Strategic Priorities

1. **For Policy Makers:**
    - Require interoperability standards in NEVI funding
    - Shift to utilization-based incentives
    - Protect rural charging access
2. **For Network Operators:**

- Phase out CHAdeMO infrastructure
- Optimize existing high-traffic sites
- Accelerate NACS adapter deployment

3. **For Businesses:**
    - Co-locate amenities with DC fast charging
    - Target emerging markets (suburban expansion zones)
    - Focus on utilization metrics over station counts
4. **For Consumers:**
    - Prioritize vehicles with CCS or NACS
    - Expect continued infrastructure expansion (but slower growth)
    - Advocate for pricing transparency

Next Steps for Deeper Analysis

Recommended Extensions

1. **Utilization Analysis:**
    - Obtain transaction data to calculate capacity utilization rates
    - Identify underperforming stations for optimization or closure
    - Correlate utilization with station size, location, network
2. **Economic Modeling:**
    - Link to EV registration data to model demand-supply equilibrium
    - Forecast break-even utilization by station type
    - Assess NEVI ROI and subsidy effectiveness
3. **Pricing Competitiveness:**
    - Comprehensive pricing data collection (currently 11% complete)
    - Cross-network price comparison
    - Demand elasticity analysis
4. **Predictive Modeling:**
    - Forecast infrastructure needs 2026-2030

- Optimal site selection using ML (demographic, traffic, competitor data)
- Scenario modeling: NACS universal adoption, EV penetration rates

5. **Equity Analysis:**
    - Map charging access to disadvantaged communities
    - Assess rural vs. urban infrastructure parity
    - Evaluate NEVI "Justice40" compliance

Data Files Generated

1. `station_level_summary.csv` — Aggregated station-level data (355 stations)
2. `connector_type_summary.csv` — Connector technology statistics
3. `nc_ev_charging_units_comprehensive_analysis.png` — 6-panel visualization
4. `nc_ev_charging_detailed_analysis.png` — 4-panel deep-dive visualization

Report Prepared By: Advanced Business Analytics Team
Date: December 29, 2025
Dataset: `alt_fuel_stations_ev_charging_units.csv` (1,725 rows × 85 columns)
Analysis Tools: Python (pandas, matplotlib, seaborn), SAS-equivalent methodologies

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